UCC 103 – PRINCIPLES OF COMPUTING
Chapter 11: Spreadsheets Application

Objectives
At the end of the chapter the learner shall be able to;

- Explain the different Spreadsheet terms
- Create a new spreadsheet and open an existing one
- Input text, numbers and simple formulae
- Employ simple functions such as SUM, AVERAGE
- Create and modify charts/graphs to illustrate data

11.1. Introduction

Spreadsheets are application packages used for manipulation of figures. A spreadsheet usually consists of a series of rows and columns. The figures or text are inserted into cells. Examples include Ms Excel, Lotus 1-2-3, Supercalc, Quattro Pro. From the examples given, we shall tackle Ms Excel.

A spreadsheet program uses worksheets and workbooks. Spreadsheets allow you to organize information in rows and tables (which create cells), with the added automatic mathematics. Spreadsheets have been used for many years in business to keep track of expenses and other calculations. Excel numerical data in a spreadsheet can be converted easily into a chart for graphical presentation of the data.

11.1.1. Terminologies

1) Worksheet – Excel's term for an electronic spreadsheet
2) Cell address – cells are "containers" that are capable of holding data. They are arranged in a series of lettered and numbered columns and rows. The cell address is also displayed in the Name box at the left end of the formula bar.
3) Workbook – multiple related worksheets.
4) Active cell – a cell with a dark border (called cell selector) around it. It's the cell that you have selected to enter or edit data.
5) Column heading – the lettered boxes across the top of the Excel's workspace. Clicking one selects an entire column of cells, which you can format or move as a whole.
6) Row heading – the numbered boxes down the side of the Excel's workspace. Clicking one selects an entire row of cells, which you can format or move as a whole.
7) Sheet tabs – click on these to move from one worksheet in a workbook to another.
8) Name box – the box at the far-left end of the Formula bar that holds the address of the cell or cell range currently selected in Excel.
9) Fill handle - the small black square in the corner of the selection. When you point to the fill handle, the pointer changes to a black cross. To copy contents to adjacent cells or to fill in a series such as dates, drag the fill handle. To display a shortcut menu that contains fill options, hold down the right mouse button as you drag the fill handle.

11.1.2. Characteristics of a good spreadsheet software

1) Allows presentation of bulky tabular data

2) Contains formulae and other functions that make complex mathematical operations easy to handle
3) Contains database tools that allow links to other applications for information processing.
4) Allows the user to represent information graphically.
5) Allows good organization of information within a given file (workbook).

11.1.3. Applications of Spreadsheets

1) Can be used to record sales, produce invoices and compile statements.
2) Researchers can compile and analyze their results.
3) Teachers can compile their students' marks and produce overall results.
4) Clerks and secretaries can easily create tables of figures and manipulate.
5) Excel can also be used to visualize data

11.2. Microsoft Excel 2010

Starting Excel

Option 1

- Click the start button
- Click All programs
- Select Microsoft Office menu
- Select and click Microsoft Excel menu item

Option 2

Click the excel button on the Microsoft shortcut if only if the option available

11.2.1. The Excel Window

Window

When you work in Excel, you use workbook files to hold your information. Each workbook consists of several worksheets made up rows and columns of information. A workbook therefore allows you to organize various kinks of related information in a single file (or workbook)

Worksheet – Excel 2010

A worksheet is one sheet in an Excel workbook. Each worksheet consists of **256 columns** and **65,536 rows**.

Excel 2016 has total **columns 16,384**, A to XFD and total rows **1 - 10,48,576,**

Columns

Columns are the vertical divisions of a worksheet that are identified by letters. The columns begin with A and proceed through the alphabet. The 27th column is AA followed by AB, AC, and this convention for naming continues through the entire alphabet until you end up with the last column (column 256) which is designated IV.

Rows

Rows are the horizontal divisions of a worksheet and are identified by numbers.

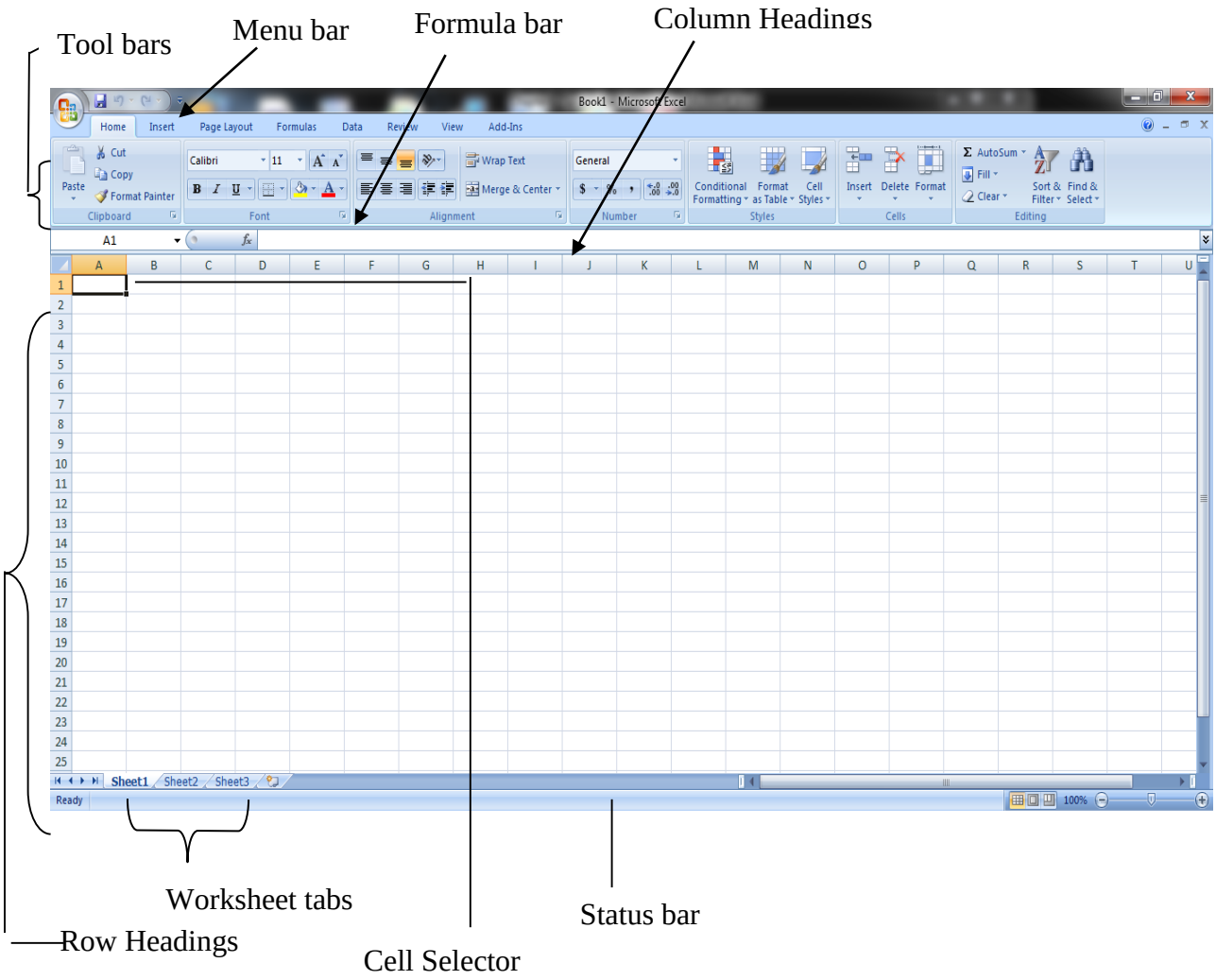
Cells

A cell is the intersection of a row and a column. Each cell has an address that consists of the column letter and row number (A1, B3, C5 and so on).

Each cell is capable of containing different types of information e.g. text, number, times, formulas. Excel data basically comes in two varieties: **labels and values.**

- A ***label*** is a text entry consisting of alphanumeric characters. It is called a label because it typically provides descriptive information such as the name of a place, person, e.t.c. A label has no numerical significance in Excel.
- A ***value*** is data that has numerical significance. These include numbers, dates and times that you enter on your worksheet. Values can be acted on by formulas and functions.

The figure below shows the elements of an Excel window.



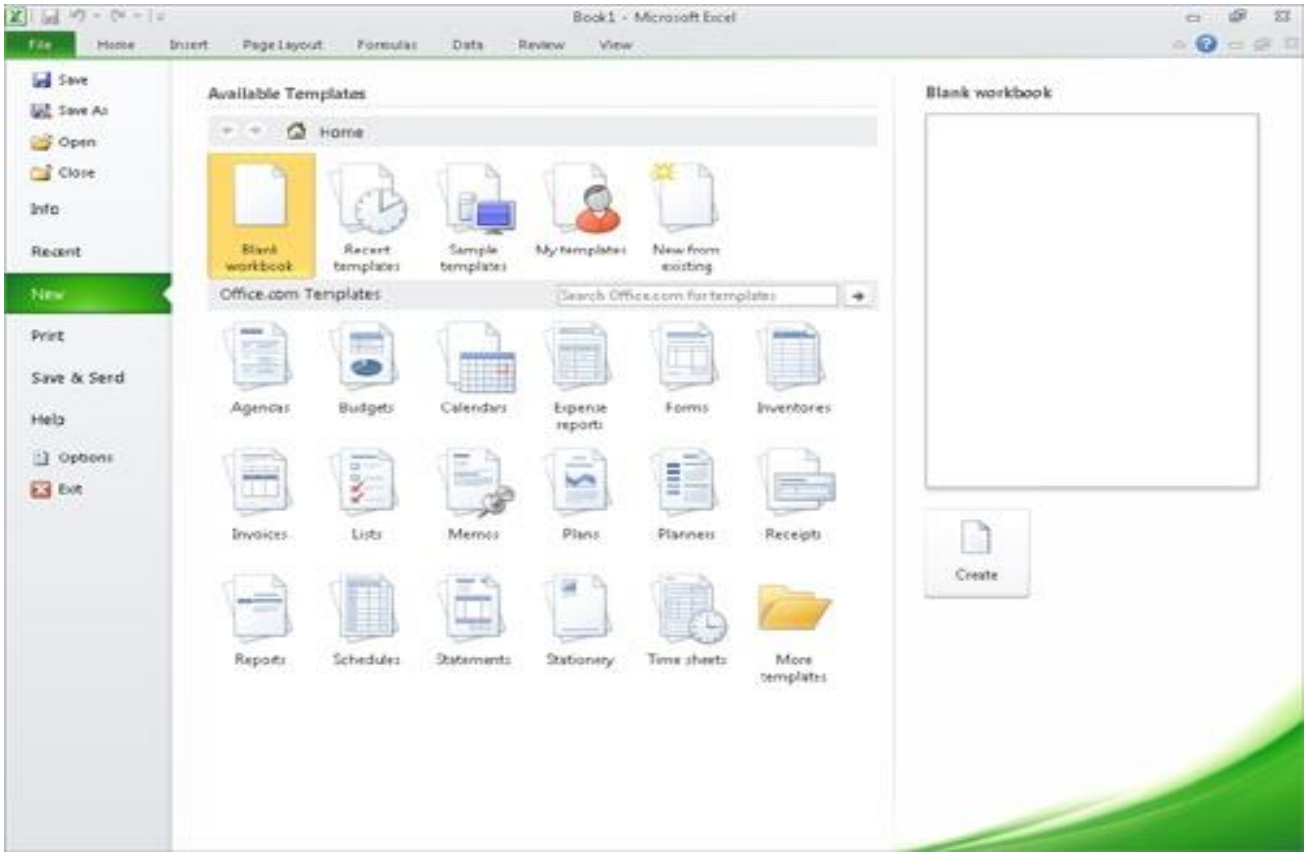
Element	Description
Formula bar / Function bar	When you enter information into a cell, it appears in the Formula bar. You can use the formula bar to edit the data later. The cell's location also appears.
Column Headings	The letters across the top of the worksheet, which identify the columns in the worksheet.
Row Headings	The numbers down the side of the worksheet, which Identify the rows in the worksheet.
Cell Selector	The dark outline that indicates the active cell. It Highlights the cell you are currently working in.
Worksheet tabs	These tabs help you move from worksheet to worksheet within the

	workbook. The active Worksheet is displayed in bold.
Active cell	It indicates the cell in which the typed data will be entered. It is also known as the current cell.

11.2.2. Create a new workbook

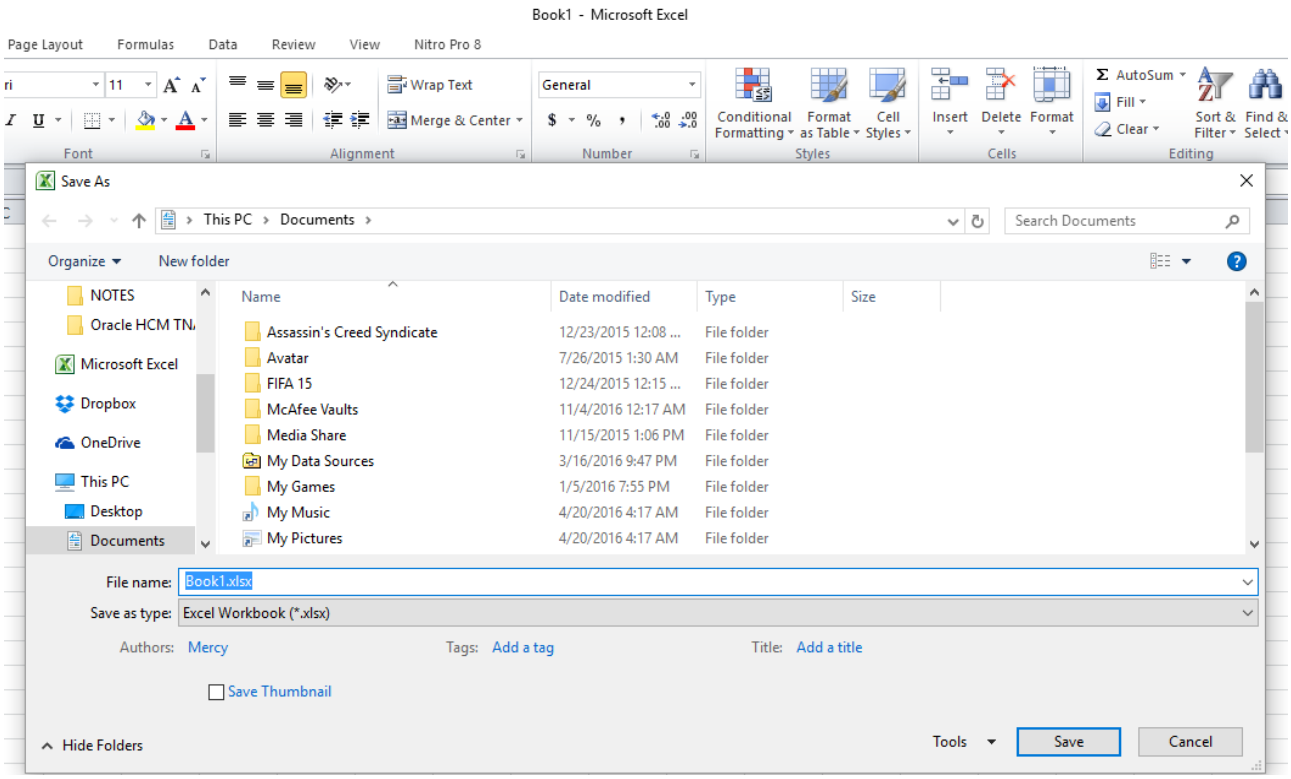
Option 1

1. Click the File menu/office button, then click New on the menu
2. Select Blank workbook and click create.



11.2.3. Saving a workbook

1. Click File on the Excel Menu Ribbon
2. Select **Save As** or **Save** as Appropriate
    - Save As allows you to: Change the name, Change the File format and storage location of the file
    - Save replaces the old version of the file using the same file name and in the same location
3. Using **Save As** options



- Type the File Name
- Select the Storage location in memory
- Click the Save as Type to select alternative file format or version for the file

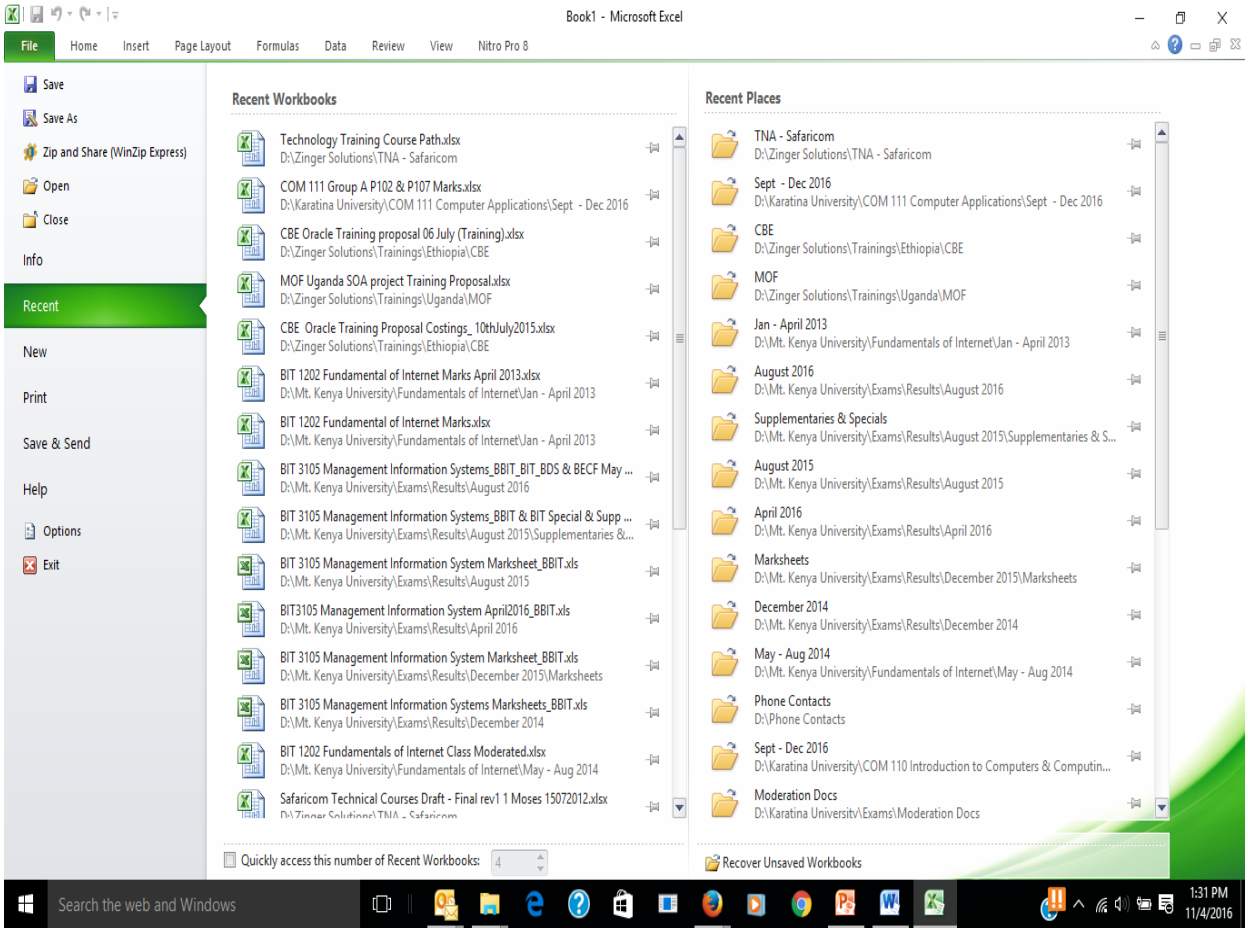
4. Click Save.

NB

The first time you click ‘save as’ from the file menu, the Documents library folder is displayed by default as the folder to save in.

11.2.4. Opening a Recent file

1) Click file menu
2) Click recent files
    - File recently opened will be displayed
3) Select and click the required file to open it



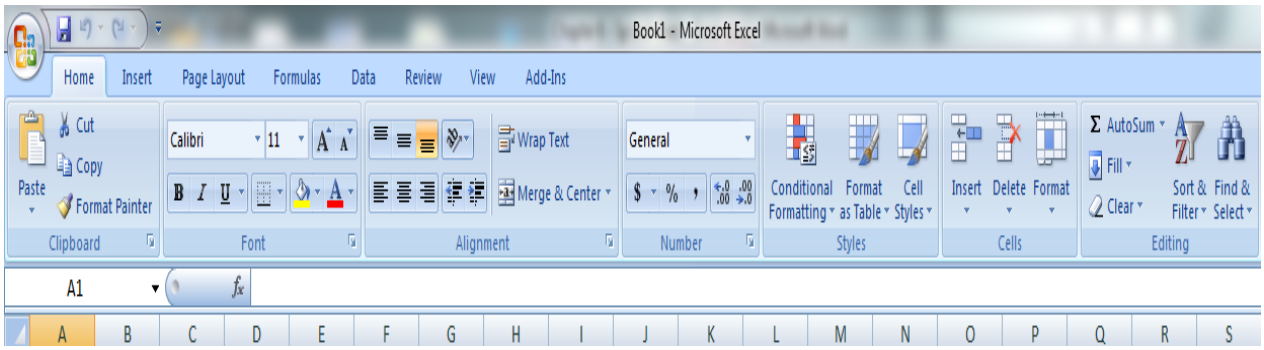
11.2.5. Close a Workbook/ Excel

1) To Close current workbook only:
   - Select File
   - Select Close.
2) To Exit: (Close workbook and Exit from Excel)
   - Select File
   - Select Exit

11.3. Excel Menus – 2010 Version

Home Menu

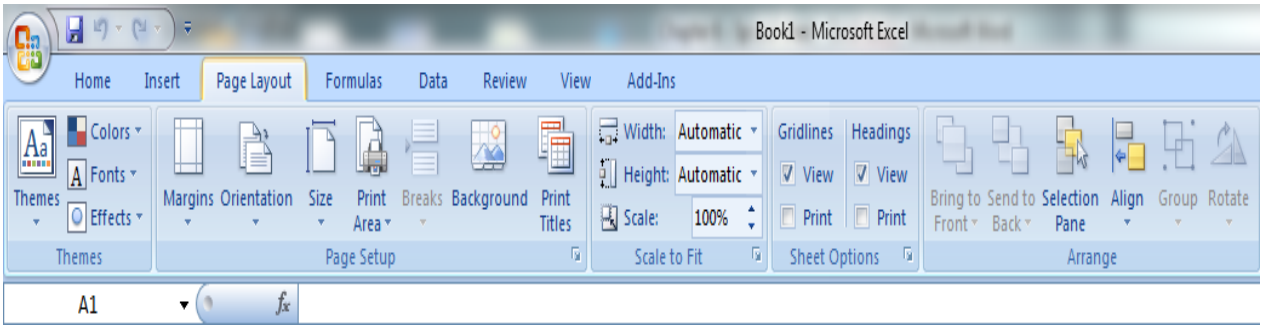
Allows access routine commands used in Excel worksheets manipulation



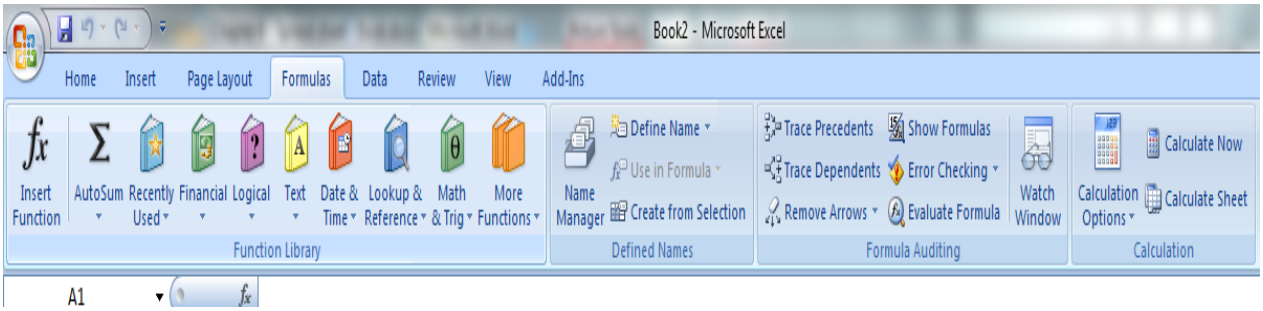
Insert Menu
Allows access to Insert command / tools



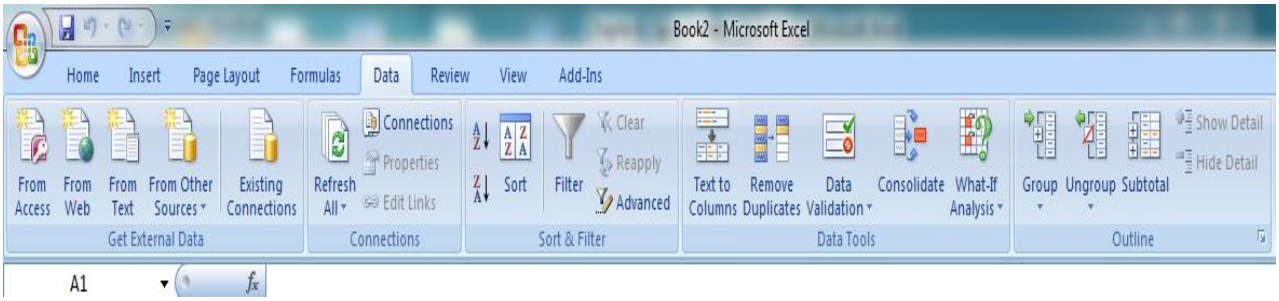
Page Layout Menu
Used to access Page layout formatting commands / tools



Formulas Menu
Allows access to the function command

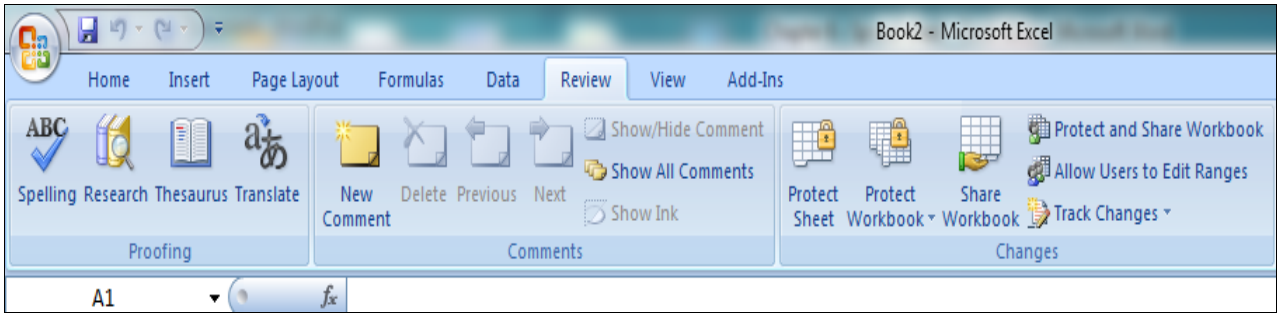


Data Menu
Access to the Data manipulation commands/tools

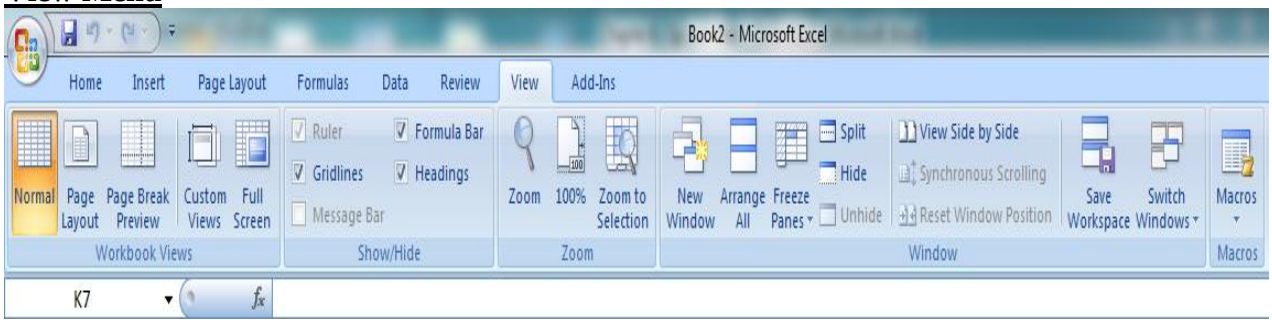


Review Menu

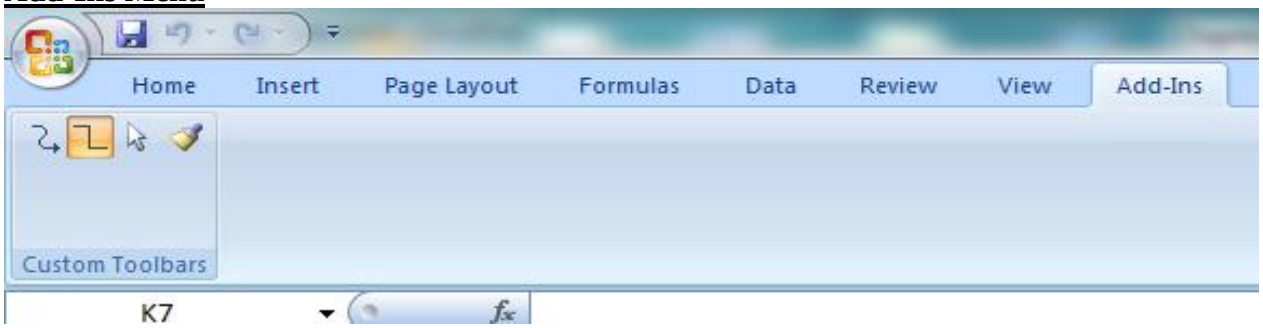
Provides Access to Document review command



View Menu



Add-Ins Menu



11.4. Working with Workbooks and Worksheets

11.4.1. Navigating in a Workbook

At the bottom of the workbook window (where the sheet tabs are located) are a number of controls you can use to move from worksheet to worksheet within a workbook. Use the four tab scrolling buttons in the lower left corner only when you have more worksheets in a workbook than can be displayed at once. You can drag the tab split box to change the number of the sheet tabs displayed. To reset the tab display, simply double-click the tab split box.

You can also navigate a worksheet using the arrows to the right of your keypad — up, down, left, or right. You can always use point and click with your mouse.

11.4.2. Moving around in a worksheet

To move between cells on a worksheet, click any cell or use the arrow keys. When you move to a cell, it becomes the active cell (the selected cell in which data is entered when you begin typing.

Only one cell is active at a time. A heavy border bound the active cell). To see the different area of the sheet, use the scroll bars.

11.4.3. Switch to another sheet in a workbook

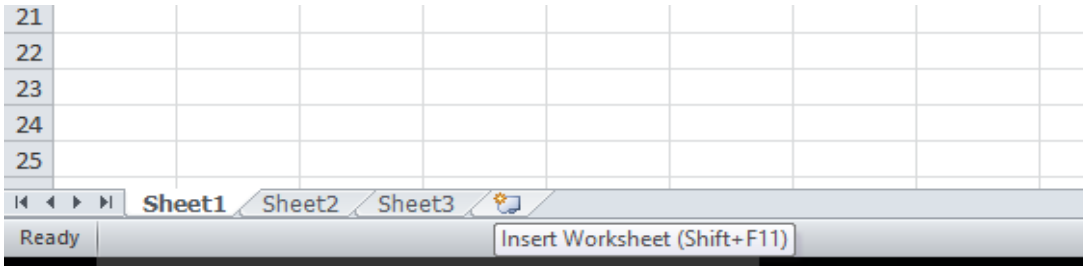
Click the sheet tab

(A tab near the button of a workbook window that displays the name of a sheet).

- To display a shortcut menu, click a tab with the right mouse button.
- To scroll through the sheet tabs, use the tab scrolling buttons to the left of the tabs)

11.4.4. Insert a new worksheet

1. Click the Insert Worksheet Button



11.4.5. Delete sheets from a workbook

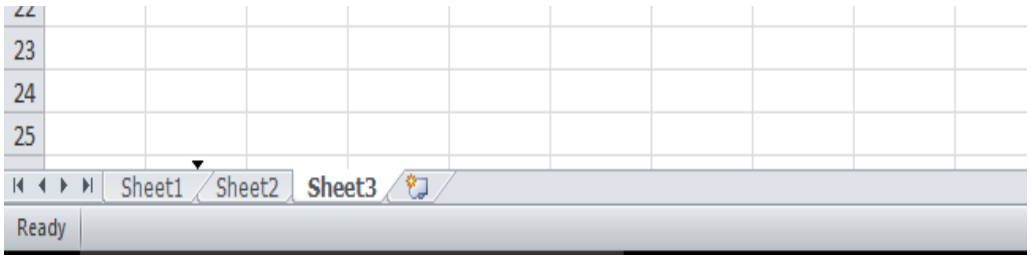
1. Click the worksheet name
2. Right Click, then Click Delete.

11.4.6. Rename a sheet

1. Click the worksheet name
2. Right click, then click rename.
3. Type the new name over the current name.

11.4.7. Moving a Worksheet

1. Click the relevant worksheet
2. While holding the Left Mouse Click button Move to the position to move the selected worksheet
3. Release the Button

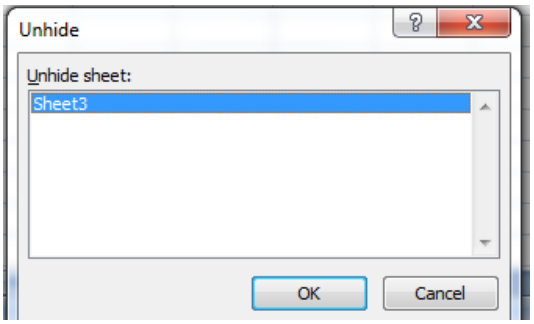


11.4.8. Hide a sheet

1. Click the worksheet name want to hide.
2. Right click,
3. Click hide.

11.4.9. Display a hidden sheet

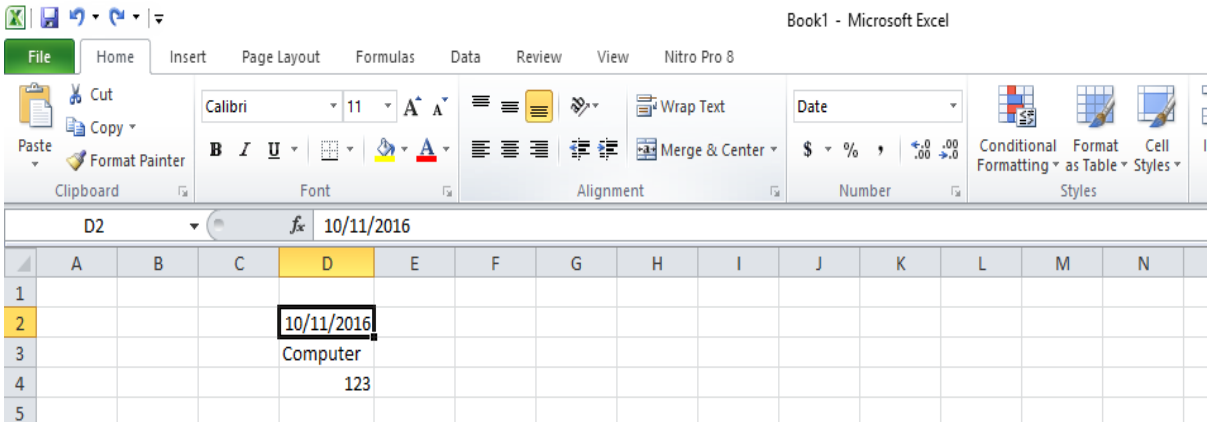
1. Click any worksheet name
2. Right Click, and click Unhide
3. Choose the worksheet to unhide
4. Click Ok.



11.5. Entering data

Enter numbers, text, date, or time

1. Click the cell where you want to enter data.
2. Type the data and press ENTER or TAB.
    - Text is left aligned
    - Numbers are right aligned
    - Date is Left Aligned



NB. When the sell with data is selected, the type of data is automatically displayed in the Number Sub-menu of the Home Menu

Enter the same data into several cells at once

1. Highlight the cells where you want to enter data.
   The cells can be adjacent or nonadjacent
2. Type the data and press CTRL+ENTER

Fill in a series of numbers, dates or other items

1. Select the first cell in the range you want to fill and enter the starting value for the series.

To increment the series by a specified amount, select the next cell in the range and enter the next item in the series. The difference between the two starting items determines the amount by which the series is incremented.

2. Select the cell or cells that contain the starting values.
3. Drag the fill handle over the range you want to fill.
   - To fill in increasing order, drag down or to the right.
   - To fill in decreasing order, drag up or to the left.

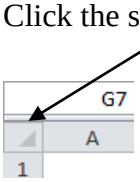
Cancel or undo an entry

1. To cancel an entry before you press ENTER, press ESC.
2. To undo a complete entry, click Undo button on the standard tool bar.

Tips on entering numbers

1. To avoid entering a fraction as a date, precede fractions with a 0 (zero); for example, type0 ½ (there is a space between zero and ½)
2. Precede negative numbers with a minus sign (-), or enclose the numbers in parentheses ().

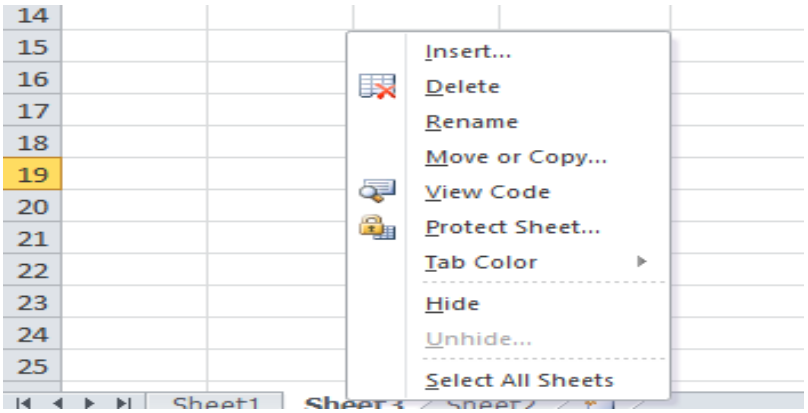
Select cells, ranges, rows and columns

To select	**Do this**
A single cell	Click the cell, or press the arrow keys to move to the cell
A Range of cells/ Block of cells	Click the first cell of the range and then drag to the last cell.
All cells on a worksheet	Click the select All button  Click Ctr+A
Nonadjacent cells or cell ranges	Select the first cell or range of cells, and then hold down CTRL and select the other cells or ranges.
A large range of cells	Click the first cell in the range, and then hold down SHIFT key and click the last cell range
An entire row	Click the row heading.
An entire column	Click the column heading.
Adjacent rows or columns	Drag across the row or column headings
Non-adjacent rows or columns.	Select the first row or column, and then hold down CTRL and select the other rows or columns

11.6. Select sheets in a Workbook

If you select more than one sheet, Microsoft Excel repeats the changes you make to the active sheet on all other selected sheets. These changes may replace data on other sheets.

To select	***Do this***
A single sheet	Click the sheet tab. Press Ctr +A
Two or more adjacent sheets	Click the tab for the first sheet and then hold down SHIFT and click the tab for the last sheet.
Two or more nonadjacent sheets	Click the tab for the first sheet and then hold down CTRL and click the tabs for the other sheets.
All sheets in a workbook	Right click a sheet tab and then click Select All Sheets on the shortcut menu.



- To cancel a selection of multiple sheets in a workbook, click any unselected sheet.
- If no unselected sheet is visible, right click the tab of a selected sheet then click Ungroup Sheets on the shortcut menu.

11.7. Formatting a Worksheet

11.7.1. Change the size, font, colour, or other text format

You can specify a font, font size and font color by clicking buttons on the Formatting toolbar.

Change the font or font size

1. Select whole cells or the specific text in a single cell that you want to format.
2. In the Font box, click the font you want.
3. In the Font size box, click the font size you want (on the formatting toolbar)

Make selected text or numbers bold, italic or underlined

1. Select whole cells or the specific text in a single cell that you want to format.
2. On the formatting toolbar, click a button for the format you want.

To make text	***Click***
Bold	**B**
Italic	***I***

Underlined **U**

Change the text color

1. Select whole cells or the specific text in a single cell that you want to format.
2. To apply the recently most selected color, click Font Color 
3. To apply a different color, click the arrow next to Font Color Icon and then click a color on the palette.

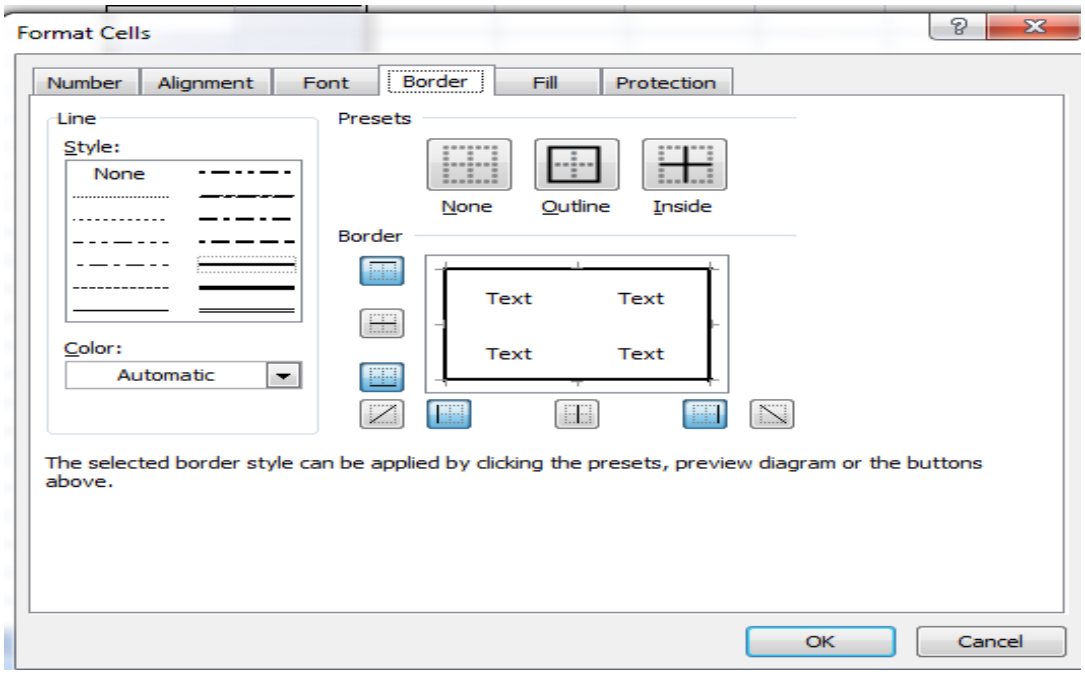
11.7.2. Applying borders

To apply border styles

1. Click Cells and on the Home Menu, click the Borders tab
2. Select the border type from the list

To change the line style of an existing border

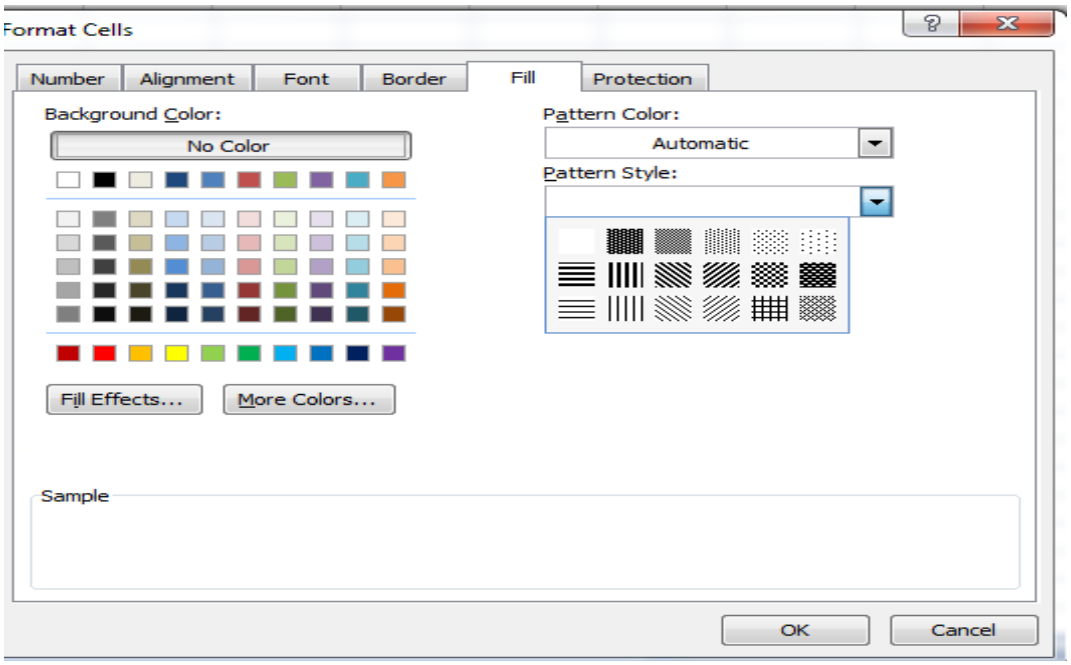
1. Select the cells on which the border is displayed.
2. Click the Borders tab on the Home Menu
3. Select More Borders tab
4. Select the line style
5. Click Ok



11.7.3. To shade cells with patterns

1. Select the cells on which the border is displayed.
2. Click the Borders tab on the Home Menu
3. Select More Borders tab
4. Click Fill tab
5. Click to open the pattern style selection box

6. Select the pattern style
7. Click Ok



Add background patterns to an entire sheet

1. Click the sheet to which you want to add a background pattern
2. On the page layout Menu Click the Background tab
3. Select the graphics file to use for the background pattern

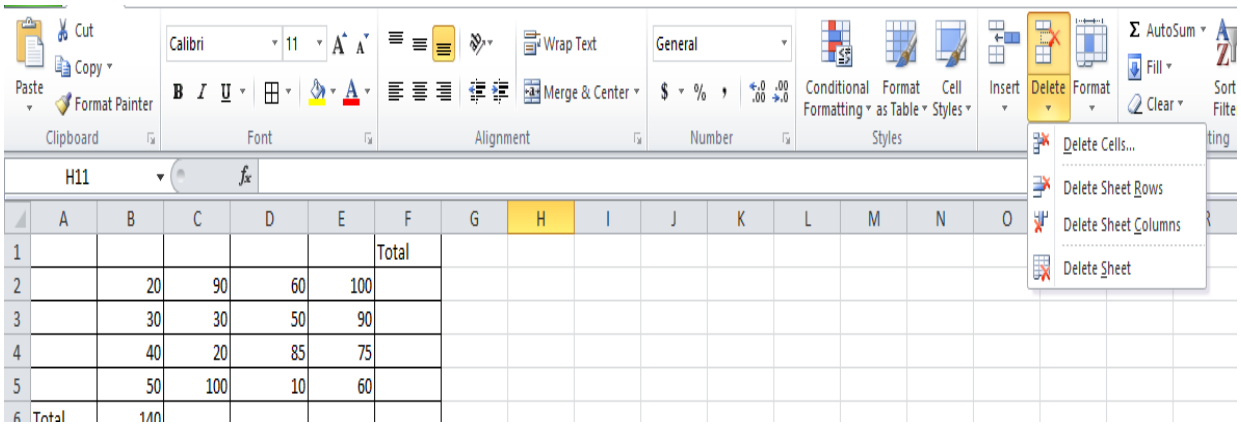
The selected graphic is repeated to fill the sheet. You can apply solid color shading to cells that contain data.

11.8. Working with Columns Rows and Cells

Clear or delete cells, rows or columns

- When you delete cells, Microsoft Excel removes them from the worksheet and shifts the surrounding cells to fill the space.
- When you clear cells, you remove the cell contents (formulas and data), formats or comments but leave the blank cells on the worksheet.

1. Select the cells, rows or columns that you want to clear.
2. On the Cell sub-menu of the Home Menu,
3. Click Delete
4. Choose the appropriate command



Undo mistakes

1. To undo recent actions one at a time, click Undo on the edit menu.
2. To undo several actions at once, click the arrow next to Undo button on the standard toolbar and select from the list. Microsoft Excel reverses the selected action and all actions above it.
3. To undo several actions at once, click the arrow next to Undo button on the standard toolbar and select from the list. Microsoft Excel reverses the selected action and all actions above it.

Insert cells, rows or columns

You can insert blank cells, rows and columns and fill them with data.

Insert blank cells

1. Select a range of existing cells where you want to insert the new blank cells.
2. On the Cell sub-menu of the Home Menu,
3. Select Insert cell

.

Insert rows

1. To insert a single row, click a cell in the row immediately below where you want the new row. For example, to insert a new row above Row 5,click a cell in Row 5.
   To insert multiple rows, select rows immediately below where you want the new rows, select the same number of rows you want to insert.
2. On the Home Menu, click insert.
   The application to insert rows above the selection

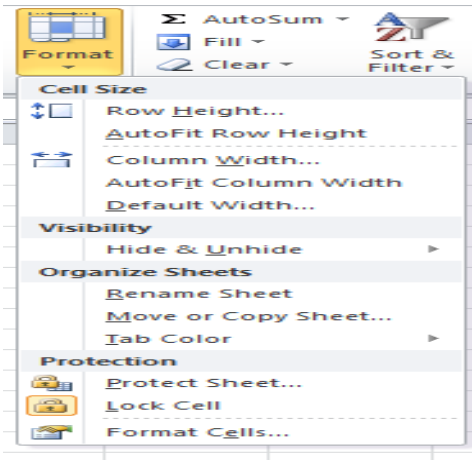
Insert columns

1. To insert a single column, click a cell in the column immediately to the right of where you want to insert the new column. For example, to insert a new column to the left of column B, click a cell in column B
2. To insert multiple columns, select columns immediately to the right of where you want to insert the new columns. Select the same number of columns as you want to insert.
3. On the Home Menu, Click insert.

Re-Sizing Columns and Rows Heights & Width

1. Select the rows or Column
2. On the Cell sub-menu on the Home Menu

3. Select Format menu
4. Choose the appropriate cell Sizing option



OR

For Column

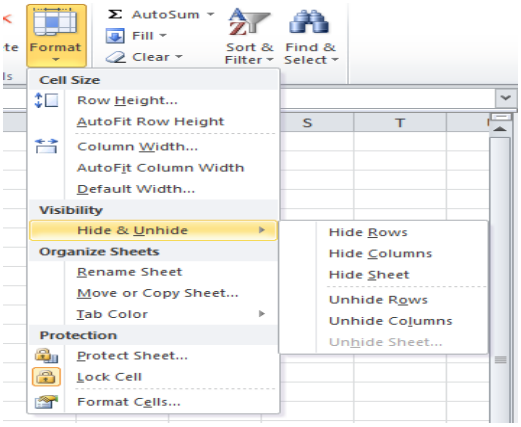
1. Select the Colum
2. Drag the boundary on the right side of the column heading until the column is the width you want.
3. To change the column width for multiple columns, select the columns you want to change. Then drag a boundary at the right of a selected column heading.
4. To change the column width for all columns on the worksheet, click the Select All button (at the left edge of the borders) and then drag the boundary of any column heading.

Row

1. Select the row
2. Drag the boundary Up or down of the row heading until the row is the height you want.
3. To change the rows height for multiple rows, select the rows you want to change. Then drag a boundary up or down of a selected row heading.

Hiding and Unhide Columns and Rows

1. Select the respective Columns or Rows
2. On the Cell sub-menu on the Home Menu
3. Select Format menu
4. Select Hide & Unhide menu
5. Select the Appropriate option



You can adjust the width of columns and the height of rows. You can also define the default width of columns for a worksheet. Defining the default column width adjusts all columns to the same width except columns that have previously been changed.

Change column width

Change row height

Using different options

1. Drag the boundary below the row heading until the row is the height you want.
2. To change the row height for multiple rows, select the rows you want to change then drag a boundary below a selected row heading.
3. To change the row height for all rows on the worksheet, click the Select All button (at the left edge of the borders) and then drag the boundary below any row heading.
4. To make the row height fit the contents, double-click the boundary below the row heading.

The Fill Handle

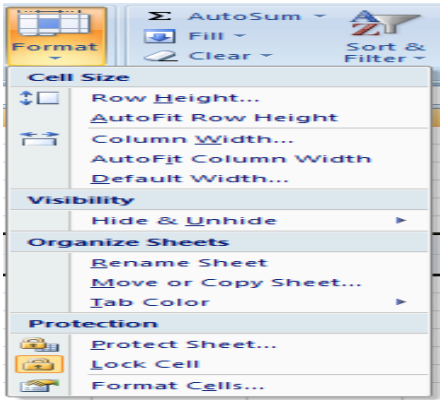
The fill handle enables you to extend a series. It is also used for copying formulas.

Procedure

1. Position the mouse pointer right on the block like mark in the bottom right corner of the active cell.
2. Click and drag to extend a series.

Format Command

1. Select the rows or columns
2. On the Home menu click to open the Format sub-menu
3. Select the type of formatting required
4. Specify the row height or Column Width
5. Click Ok.



11.9. Excel Operators

Operators are signs or symbols which specify the type of a calculation that you may perform in the elements of a formula.

There are four different types of calculation operators i.e.

1. Arithmetic
2. Comparison
3. Text and
4. Reference

11.9.1. Arithmetic Operators

They perform basic mathematical operations e.g. addition, subtraction, multiplication and division.

Operator	*Performs*	*Sample Formula*	*Result*
^	Exponentiation	=A1^3	Enters the result of raising the value in Cell A1 to the third power.
+	Addition	=B1+B2	Enters the total of the values in cells B1 and B2.
-	Subtraction	=B1-B2	Subtracts the value in cell B2 from the value in cell B1.
*	Multiplication	=A1*B1	Multiplies the value cell A1 by cell B1.
/	Division	=A1/B1	Divides the value in cell A1 by the value in cell B1.

11.9.2. Comparison Operators

They compare two values then produce a logical value i.e. TRUE or FALSE.

OPERATOR	*EXAMPLE*
= (Equal to)	A1=B1
> (Greater than)	A1>B1
< (Less than)	A1<B1
>= (Greater or equal to)	A1>=B1
<= (Less or equal to)	A1<=B1
<> (Not equal to)	A1<>B1

11.9.3. Reference Operators

Reference operators refer to a cell or a group of cells. There are two main types of reference operators:

1. Range: - **: (FULL COLON)**

- Refers to all of the cells between and including the reference.
- Consists of two cell addresses separated by a colon.
  Example 'A1:A3' includes cells A1, A2, and A3.
  'A1:C3' includes cells A1, A2, A3, B1, B2, B3, C1, C2, and C3.

2. Union:- **, (COMMA)**

- Includes two or more references.
- Consists of two or more cell addresses separated by a comma.
  Example 'A7, B8, C9' refers to cells A7, B8, and C9.
  'A7, B8:D9, E4' refers to cells A7, B8, B9, C8, C9, D8, D9 and E4.
  =SUM(B5:B15, D5:D1)

11.10. FORMULAS AND FUNCTIONS

The ability to perform calculations is one of the purposes of using a spreadsheet application. Some examples of the types of calculations that can be done are:

1) totals
2) subtotals
3) average
4) standard deviation

In Excel, the calculation can be specified using either a **formula** or a **function**.

1. Formulas are self-defined instructions for performing calculations.
2. In contrast, functions are pre-defined formulas that come with Excel.

11.10.1. Formula

A formula is an equation that analyses data in a worksheet. Formulas perform operations e.g. addition, multiplication, comparison etc. They can refer to other cells on the same worksheet as well as other sheets in the same workbook or even in other workbooks.

When using a formula in your spreadsheet, (a cell containing a formula that references other cells), the sum will automatically change as other cell values referenced in the formula change. This feature is very valuable when editing or adding information to your worksheet. You do not have to remember to update other cells that rely on that cells information.

A formula always begins with an equal sign (=) followed by some combination of numbers, text, cell references, and operators. If a formula is entered incorrectly, an ERROR IN FORMULA message will appear.
e.g**.=10-5**. The result of the formula is then displayed in the cell.

You can use **parentheses** to change the **syntax** (structure or order of elements) e.g. in the formula =5+2*3 Excel carries multiplication first. If the parentheses are used, the syntax changes.
E.g. =(5+2)*3

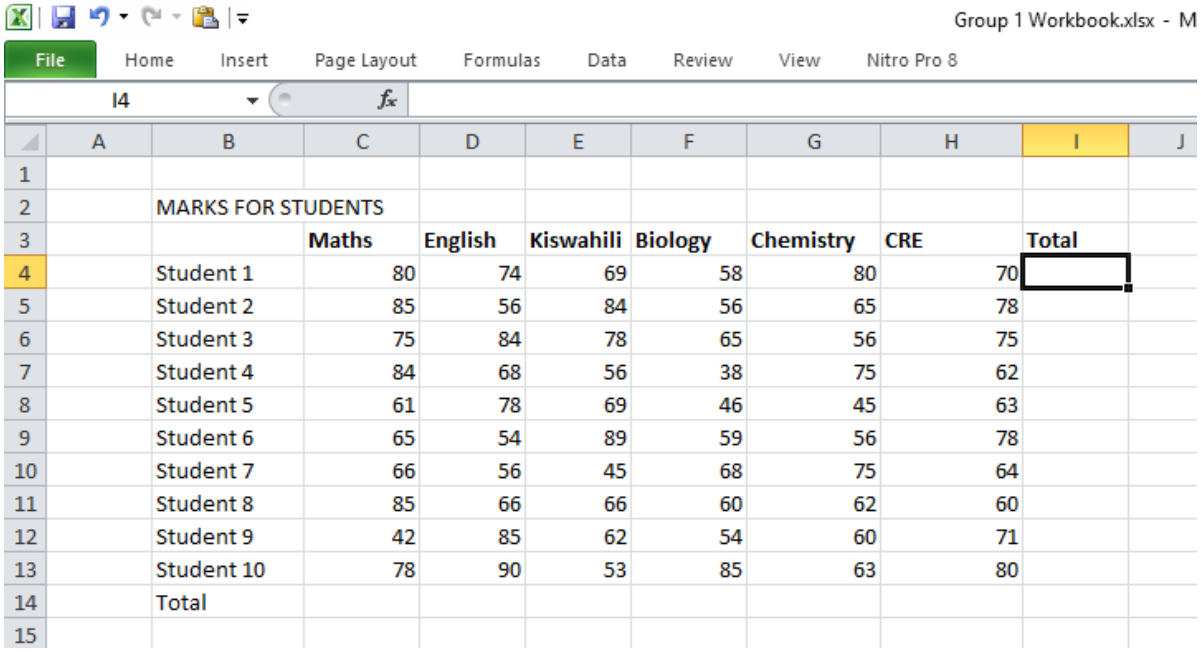
Entering Formula

After the equal sign, a formula includes the addresses of the cells whose values will be manipulated with appropriate operands placed in between. The operands are the standard arithmetic operators:

Operator	Meaning	Example
+	Addition	=A7+A9 =A1:B10
-	Subtraction	=A7-A9
*	Multiplication	=B3*D5
/	Division	=B7/C6
^	Exponents	***=A7^A9***

Example

Using the sample below



	A	B	C	D	E	F	G	H	I	J
1										
2		MARKS FOR STUDENTS								
3			**Maths**	**English**	**Kiswahili**	**Biology**	**Chemistry**	**CRE**	**Total**	
4		Student 1	80	74	69	58	80	70		
5		Student 2	85	56	84	56	65	78		
6		Student 3	75	84	78	65	56	75		
7		Student 4	84	68	56	38	75	62		
8		Student 5	61	78	69	46	45	63		
9		Student 6	65	54	89	59	56	78		
10		Student 7	66	56	45	68	75	64		
11		Student 8	85	66	66	60	62	60		
12		Student 9	42	85	62	54	60	71		
13		Student 10	78	90	53	85	63	80		
14		Total								
15										

To calculate the total marks for each student

For student 1:

1) Place the cursor at Cell I4
2) Enter the formula =(C4+D4+E4+F4+G4+H4), then press enter

If the "=" sign is missed

The text of the formula or function will be displayed in the cell if you do not use an '=' and the calculation will not be executed.

A formula always begins with an equal sign (=) followed by some combination of numbers, text, cell references, and operators. If a formula is entered incorrectly, an ERROR IN FORMULA message will appear.

11.10.2. Excel Functions

Functions are predefined formulas that perform calculations by using specific values, called arguments, in a particular order, or structure. Functions can be used to perform simple or complex

calculations. Functions differ from regular formulas in that, after the equal sign, you supply the cell addresses but not the arithmetic operators. For example, instead of typing =A1+A2+A3+A4+A5, you may type =SUM(A1:A5). (The colon indicates the range from A1 to A5.)

In this example, SUM is a function that Excel knows about. It is important to use the correct syntax and format of a function for correct results.

Common Functions

Function Name	Example	Description
SUM	=SUM(A1:B10)	Finds the Sum of cell A1 through to B10
AVERAGE	=AVERAGE(B6:D10)	Finds the average of the cells B1 through D10.
MAX	=MAX(C1:C10)	Returns the highest number from cells C1 through C10
MIN	=MIN(A2:A100)	Returns the lowest number from cells A1 through A100
SQRT	=SQRT(D10)	Finds the square root of the value in D10
TODAY	=TODAY()	Returns the current date (leave the parentheses empty).
MEDIAN	=MEDIAN(D5:D20)	Returns he middle number between D5 through to D20

Structure of a Function

The structure of a function begins with an equal sign (=), followed by the function name, an opening parenthesis, the arguments for the function separated by commas, and a closing parenthesis.

=Function_Name(arguments separated by commas)

Function name: For a list of available functions, click a cell and press SHIFT+F3. Argument: Arguments can be numbers, text, logical values such as TRUE or FALSE, arrays, error values such as #N/A, or cell references. The argument you designate must produce a valid value for that argument. Arguments can also be constants, formulas, or other functions.

Functions perform calculations by using specific values, called arguments, in a particular order called syntax. When using a function, remember the following:

1. Use an equal sign to begin the function.
2. Specify the function name.
3. Enclose all of the function's arguments within parentheses.
4. Use a comma to separate the function's individual arguments.

Selecting the Function - Function Wizard

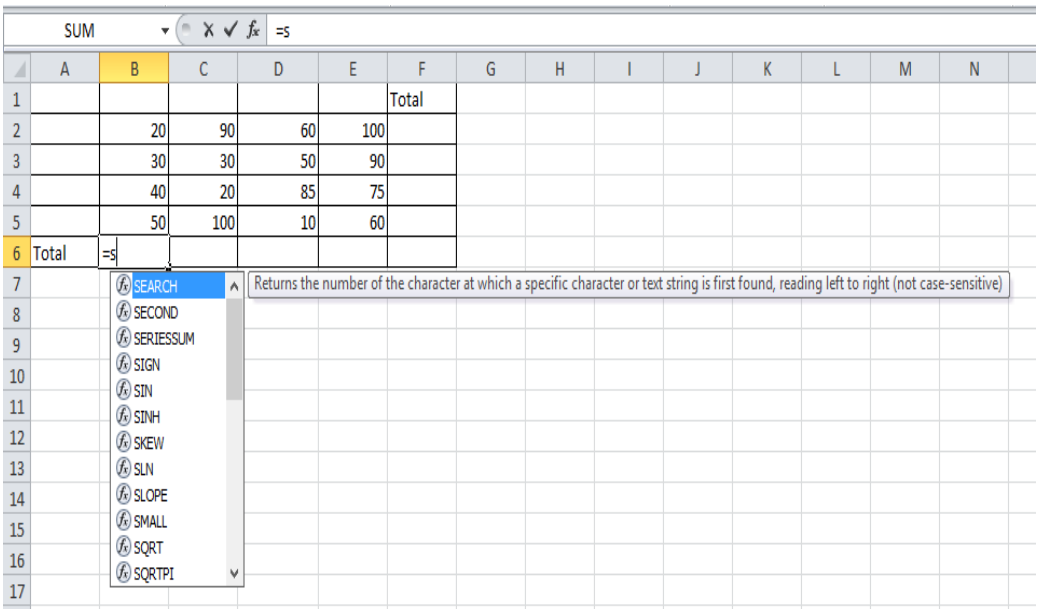
You can access all of the available functions in Excel using the Function Wizard.

- Select the cell where the function will be placed and click the Function Wizard button on the standard toolbar.

OR

1. Select the cell where the function results are to be placed

2. Type = then the First Character of the function
3. A drop-down list of ALL functions starting with the typed character are listed



4. Select the relevant function
5. Enter the function Arguments accordingly

a. THE SUM Function

The sum function sums up a range total. This function saves time e.g. instead of creating a formula = A1+B1+C1+A2+B2+C3+A3+B3+C3

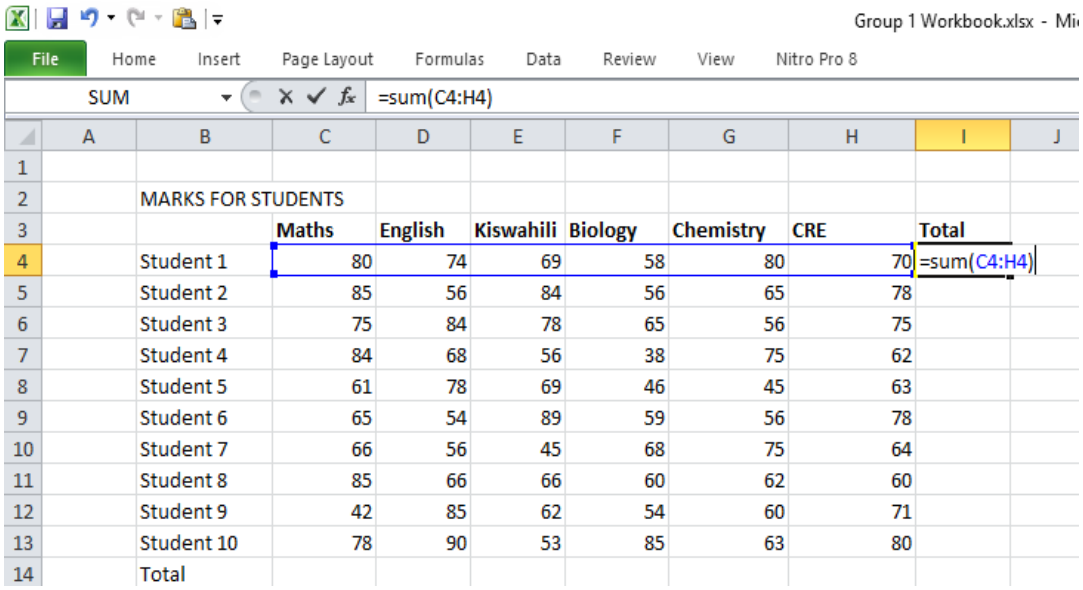
A sum function will make it easy i.e.

=SUM (A1:C3)

Please, note the reference operator **:(colon)**

Example

1) Calculating the total for marks for Student1
2) Select the Cell I4
3) Enter the formula =SUM(C4:H4)



	A	B	C	D	E	F	G	H	I	J
1										
2		MARKS FOR STUDENTS								
3			**Maths**	**English**	**Kiswahili**	**Biology**	**Chemistry**	**CRE**	**Total**	
4		Student 1	80	74	69	58	80	70	=sum(C4:H4)	
5		Student 2	85	56	84	56	65	78		
6		Student 3	75	84	78	65	56	75		
7		Student 4	84	68	56	38	75	62		
8		Student 5	61	78	69	46	45	63		
9		Student 6	65	54	89	59	56	78		
10		Student 7	66	56	45	68	75	64		
11		Student 8	85	66	66	60	62	60		
12		Student 9	42	85	62	54	60	71		
13		Student 10	78	90	53	85	63	80		
14		Total								

b. RETURN VALUES OF FUNCTIONS

1) **AVERAGE**: If cell A1 contains value 12 and B1 value 8, the function =average (A1:B1) returns 10.
2) **MODE**: This is the most frequently occurring or repetitive value in an array of data
   a. syntax e.g. =mode (10,3,4,3,5,3,7,3,4)
   b. N/B If a data set has no duplicate values, mode returns the # N/A error value.
3) **MEDIAN:** This is the middle value or the number in the middle of a set of numbers
   a. Syntax e.g. =median (2,4,6,8,10)
   Numbers in the middle e.g.
   = Median (1,2,3,4,5) equal 3
   = Median (1,2,3,4,5,6,) equals to 3.5 and this is the average of 3+4

11.11.

The various types of errors **Excel Error Values** you may encounter as you use formulas are:

Error	***Description***
• #DIV/0!	The formula is attempting to divide by zero. Check the cell references for blanks or zeros that may have resulted if you deleted a cell referenced by the formula.
• #N/A	The formula refers to a cell with a #N/A entry or a cell that contains no value This error warns you that not all the data referenced by a formula is available.
• #NAME?	Excel doesn't recognize a name you entered in a formula. Verify that all names in the formula exist and define any missing names. If applicable, verify that you used the correct function name.
• #NULL!	The formula specifies two areas that don't intersect. Check to see if you Entered the cell or range reference incorrectly. Remember to use commas (not spaces) between function arguments.
• #NUM!	There is a problem with a number used in the formula. Check for the correct use of

function arguments.

- #REF! A cell reference in the formula is incorrect. Check for changes to cell reference caused by deleting cells, rows or columns referenced by the formula.
- #VALUE! The formula contains the wrong type of argument or operator. Check for the correct syntax of the formula.

11.12. Copying Formulas and Functions

When copying the formula or function, the cell addresses will be copied in a particular way depending on how the cells are referenced.

> *A cell reference refers to a cell or a range of cells on a worksheet and can be used in a formula so that Microsoft Office Excel can find the values or data that you want that formula to calculate. In one or several formulas, you can use a cell reference to refer to: ... Data contained in different areas of a worksheet.*

Types of cell referencing

1. Relative cell reference.
2. Absolute cell reference.
3. Mixed cell reference.

Use of the **Cell reference** allows for **Automatic calculation** in case the values in the respective cells changed

Automatic recalculation. In spreadsheets, a mode in which all cells are **recalculated** whenever a value changes. Automatic recalculation ensures that the spreadsheet data is always up-to-date, but it may make working on the spreadsheet slower.

11.12.1. Using Relative cell reference

Calling cells by just their column and row labels (such as 'A1') is called a relative cell reference. When formulas or functions contain relative cell referencing and they are copied from one cell to another, Excel does not create an exact copy of the formula or function. Instead, Excel changes the cell addresses relative to the row and column they are moved to.

Example

- If a simple addition formula of '=A1+B1' in cell C1 is copied to cell C2, the formula would change to '=A2+B2' to reflect the new row.
- If a simple average function of '=AVERAGE(B2:B5)' in cell B6 is copied to cell C6, the function would change to '=AVERAGE(C2:C5)' to reflect the new column.

11.12.2. Using Absolute Cell References

An absolute cell reference refers to the same cell, no matter where the formula or function is copied. In other words, when a formula or function containing an absolute cell reference is copied to a new location, the cell reference is not adjusted.

1. To create an absolute cell reference, you will need to add dollar signs ('$') in front of both the column and row identifiers for the cell referenced – this fixes the row AND column.
2. You can add the dollar signs ('$') automatically by using the F4 keyboard shortcut – press the F4 key once after entering the cell address into the formula or function by typing or by using the point mode.

Example

- When the formula '=B7*C1' is copied from cell C7 to cell C8, the relative cell reference changes 'B7' to 'B8', but the 'C1' absolute cell reference remains unchanged.

An absolute cell reference is most often used when you want to use a constant in a formula or function.

11.12.3. Using Mixed Cell References

You use a mixed cell reference to reference a cell that is part absolute and part relative. In other words, a mixed cell reference can be used when you wish to fix only the row OR the column.
Example

- In the formula '=A$1 + $B2', the row of cell A1 is fixed and the column of cell B2 is fixed.

As with absolute cell referencing, you need to add a dollar sign ('$'), but only in front of the column OR row identifier for the cell referenced.

When using a formula in your spreadsheet, (a cell containing a formula that references other cells), the sum will automatically change as other cell values referenced in the formula change. This feature is very valuable when editing or adding information to your worksheet. You do not have to remember to update other cells that rely on that cells information.

7.14. WORKING WITH CHARTS

You can display Microsoft Excel data graphically in a chart. Charts are linked to the worksheet data they are created from and are updated when you change the worksheet data. You can create charts from cells or ranges that are not next to one another.

1. Select the cells that contain the data that you want appear in the chart.
2. On the Insert Menu, Select the Chart type from the Charts tab
3. Follow the instructions in the Chart Wizard.

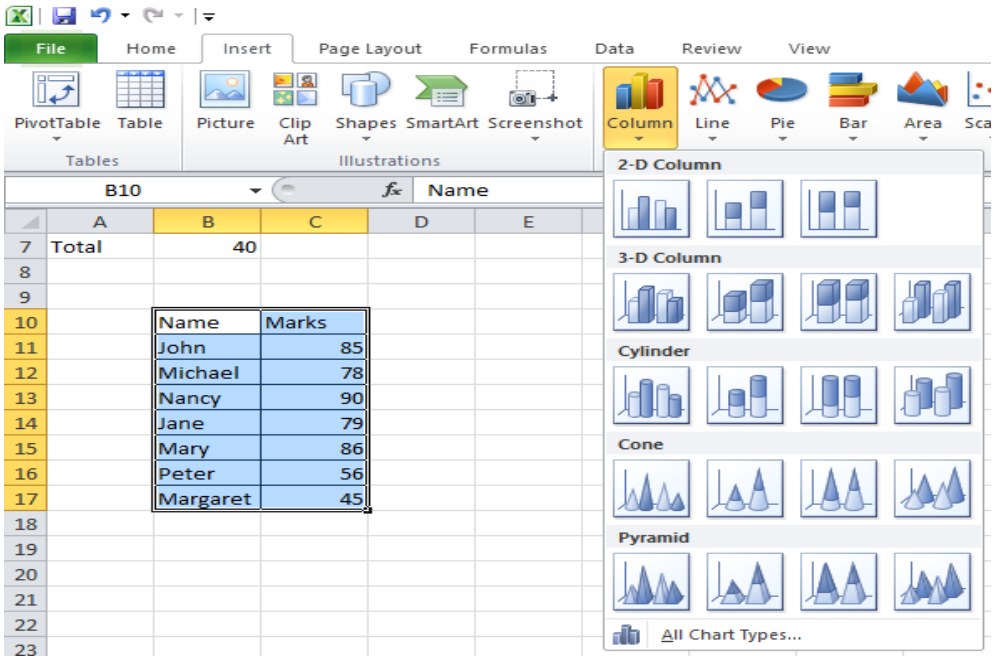
Example
Enter the following Data

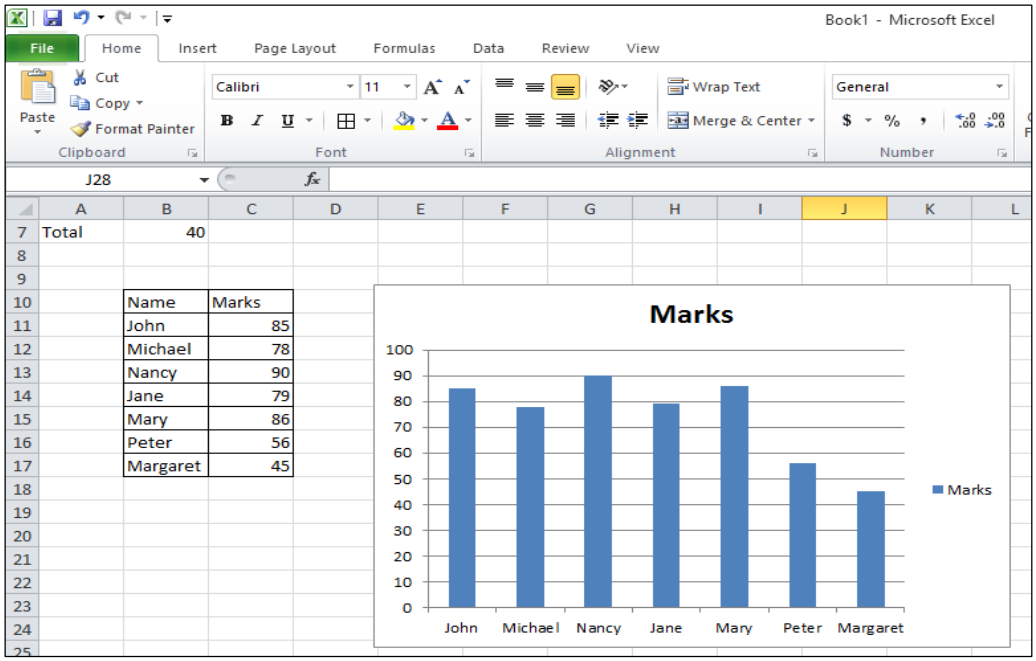
Name	Marks
John	85
Michael	78
Nancy	90
Jane	79
Mary	86
Peter	56
Margaret	45

To draw a Bar Graph

1. Highlight the Data
2. Click Insert Menu

3. Select Colum from the Charts section
4. Select the 1st option of the 2-D Column charts





Create a chart from non-adjacent selections

1. Select the first group that contains the data you want to include.
2. While holding down CTRL key, select any additional cell groups you want to include.
3. On the Insert Menu, Select the Chart type from the Charts tab
4. Follow the instructions in the Chart Wizard.

Add a text box to a chart

1. Click the chart to which you want to add a text box.
2. On the Insert Menu, click Text box Icon.
3. Click where you want the text box and then drag until the box is the size you want.
4. Type the text you want in the box. The text will wrap inside the box.
5. When you finish typing, press ESC or click outside of the text box.

Select a different chart type

1. Click the chart you want to change.
2. On the Chart menu, click Chart type.
3. On the Standard Types or Custom types tab, click the chart type you want

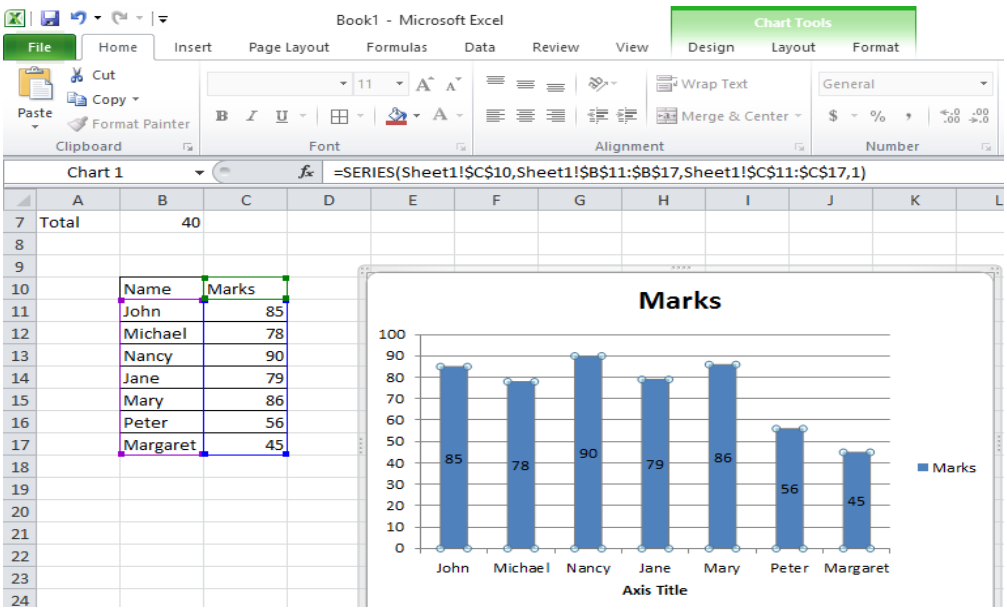
Custom chart types

1. When you create a chart or want to change the chart type, you can choose between a standard chart type and a custom chart type.
2. With a custom chart type, which is similar to a template, you can quickly change the look of your chart. Each custom chart type is based on a standard chart type and contains additional formatting and options such as a legend, gridlines, data labels, a secondary axis, colors, patterns, fills and placement choices for various chart items.

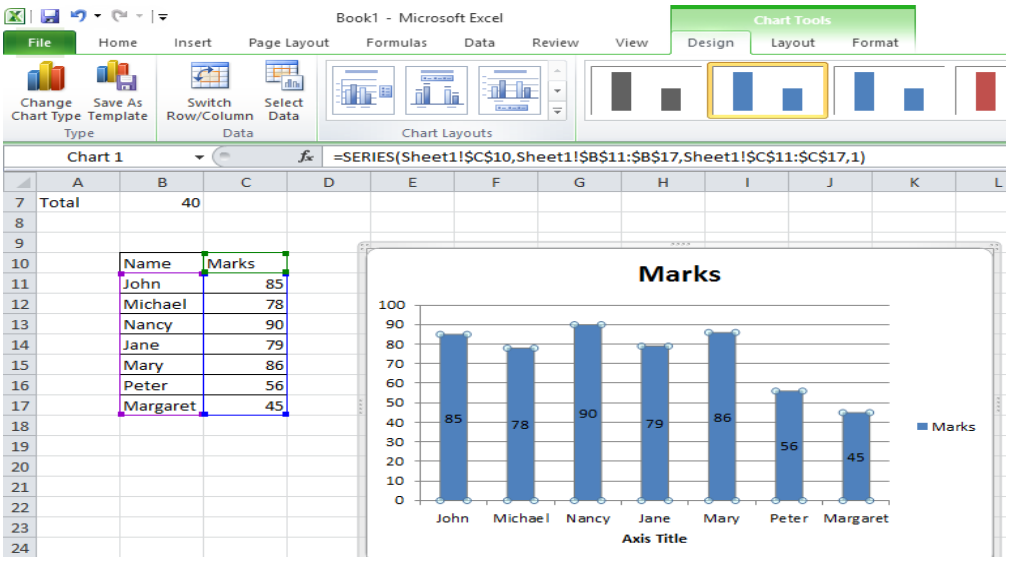
1. Click the chart you want to change.
2. On the Chart menu, click chart type.
3. On the custom Types tab, click the chart you want.

Example : To add the Axis labels

1) Click the Chart, Excel automatically highlights the Charts Tool Menu



2) Select Design sub-menu
3) Select the appropriate design from the Charts layouts



Move and resize chart items by the use of the mouse

You can use the mouse to resize and move the chart area, the plot area and the legend. Microsoft Excel automatically sizes titles to accommodate their text. You can move titles with the mouse but not resize them.

1.Click the chart item.

2.To move a chart item, point to the item and then drag it to another location.

To resize a chart item, point to a sizing handle.

When the mouse pointer changes to a double-headed arrow, drag the sizing handle until the item is the size you want.

Rotate text in a chart title or along an axis

You can rotate or "angle" text in a chart or along an axis. However, you cannot rotate legend text.

1. Click the axis or the title you want to format.
2. If you clicked an axis, click Axis on the Format menu.
3. Click the Alignment tab.
4. To rotate text, under Orientation, click a degree point or drag the indicator to the position you want.

7.15. About using a list as a database

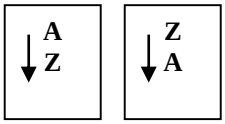
In Microsoft Excel, you can easily use a list as a database. When you perform database tasks such as finding, sorting or subtotaling data, Microsoft Excel automatically recognizes the list as a database and uses the following list elements to organize the data.

- The columns in the list are the fields in the database.
- The columns labels in the list are the fields names in the database.
- Each row in the list is a record in the database.

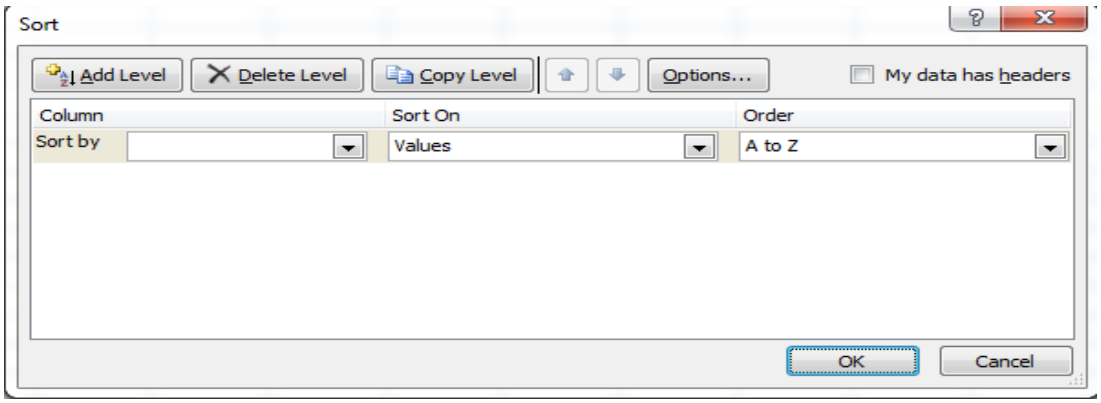
Sorting a list

You can rearrange the rows or columns of a list based on the values in the list by sorting. When you sort, Microsoft Excel rearranges rows, columns or individual cells by using the sort order that you specify. You can sort lists in ascending (1 to 9,A to Z) or descending (9 to 1,Z to A) order and sort based on the contents of one or more columns.

Sort in ascending or descending order



1. Click a cell in the column you would like data sort.
2. On the Data Menu, Click the Sort Button
3. Select the sort type
4. Click Ok.



Sort columns based on the contents of rows

1. Click a cell in the list you want to sort.
2. On the Data menu, click sort.
3. Select the Sort by Column
4. Select the Sort on value
5. Select the Sort order
6. Click Ok.

Using apply filter

Autofilter

Displays only those rows that match the value in the active cell and inserts Autofilter arrows to the right of each column label.

Show all

Displays all of the rows in a filtered list.